Sukkur Institute of Business Administration

BS – V SOFTWARE ENGINEERING

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Software Quality Engineering — Course Project Proposal

Parcel Dispatch and Tracking System

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1. Problem Statement

Courier services need an efficient and reliable way to manage parcel registration, delivery charges, staff assignments, shipment tracking, and failed deliveries. When these activities are handled manually or through disconnected records, errors such as duplicate tracking numbers, incorrect delivery charges, invalid shipment status updates, and incomplete delivery records can occur.

The proposed Parcel Dispatch and Tracking System is a web-based application designed to address these problems by centralizing parcel management and enforcing consistent business rules throughout the delivery process. The system will allow authorized staff to register parcels, calculate delivery charges, assign delivery personnel, and update shipment statuses. Customers will be able to track their parcels using a unique tracking number.

The project will emphasize Software Quality Engineering practices throughout development, including requirements analysis, software reviews, systematic testing, defect tracking, and quality measurement. The objective is to deliver not only a functional application but also a reliable, maintainable, and usable software product supported by documented quality evidence.

2. Project Objectives

The main objectives of this project are:

1. To develop a web-based system for managing parcel registration and dispatch operations.
2. To calculate delivery charges accurately based on predefined weight, distance, and delivery-type rules.
3. To generate unique tracking numbers and maintain a complete shipment history.
4. To manage delivery staff assignments and enforce valid shipment status transitions.
5. To handle failed delivery attempts and return-to-sender operations systematically.
6. To identify and document functional and non-functional requirements with measurable acceptance criteria.
7. To design and execute unit, integration, system, and user acceptance tests, where applicable.
8. To identify, classify, track, and resolve software defects through a systematic defect-management process.
9. To evaluate software quality using measurable metrics, including test pass rate, requirements test coverage, and defect retest success rate.
10. To use GitHub for version control, collaborative development, issue tracking, and documentation.

3. Project Scope

3.1 In-Scope Features

The initial version of the system will include the following functionality:

- **Parcel Registration:** Register sender and receiver information, parcel weight, origin, destination, and delivery distance.
- **Delivery Charge Calculation:** Calculate shipping charges using predefined pricing rules, including an optional express-delivery surcharge.
- **Tracking Number Generation:** Generate a unique tracking number for each registered parcel.
- **Delivery Staff Assignment:** Assign parcels to active delivery personnel and prevent invalid assignments.
- **Shipment Status Tracking:** Record shipment progress, including booking, assignment, dispatch, delivery attempts, and final delivery status.
- **Failed Delivery Management:** Record unsuccessful delivery attempts and their reasons.
- **Return-to-Sender Management:** Manage eligible parcels through the return process.
- **Customer Tracking:** Allow customers to view current parcel status and shipment history using a tracking number.
- **Quality Assurance and Testing:** Validate functionality, test boundary conditions, document defects, perform retesting, and calculate software quality metrics.

3.2 Out-of-Scope Features

To keep the project manageable within the available time, the following features will not be included in the initial version:

- Live GPS tracking and interactive maps.
- Online payment gateway integration.
- Real-time SMS or email notifications.
- Integration with external courier companies or shipping APIs.
- Advanced route optimization and automated delivery scheduling.

4. Target Users

The system will support the following user groups:

1. Administrator / Courier Clerk

Responsible for registering parcels, calculating delivery charges, assigning delivery personnel, managing shipment records, and reviewing operational information.

2. Delivery Staff

Responsible for viewing assigned parcels, recording delivery attempts, updating permitted shipment statuses, and reporting unsuccessful deliveries.

3. Customers

Able to enter a tracking number and view the current shipment status and available tracking history without accessing administrative information.

5. Proposed Technology Stack

Component	Proposed Technology	Purpose
Frontend	HTML, CSS, JavaScript, EJS	User interface, forms, pages, and dynamic server-rendered views
Backend	Node.js with Express.js	Application logic, routing, validation, and request handling
Database	MySQL	Persistent storage of users, parcels, assignments, and shipment history
Testing	Jest and Supertest	Unit and HTTP integration testing
Version Control	Git and GitHub	Source control, collaboration, branches, and issue tracking
Documentation	Markdown and project reports	Requirements, test plans, defect logs, and quality metrics

EJS (Embedded JavaScript Templates) will be used with Express.js to generate dynamic HTML pages. MySQL will store application records, while automated tests will verify the correctness of business rules and application behavior.

6. Group Members and Assigned Responsibilities

Group Member	Primary Responsibility	Assigned Tasks
Shaheer Ahmed Siddiqui	Development	Implement backend functionality, application routes, business logic, database integration, and frontend integration.
Lakesh Kumar	Testing and Evaluation (QA)	Prepare test plans and test cases, perform unit and integration testing, record and track defects, conduct regression testing, and calculate quality metrics.
Safeer Ahmed Mirani	Documentation	Prepare the project proposal, Software Requirements Specification (SRS), design documentation, test documentation, final report, and presentation materials.

All group members will participate in requirements discussions, code reviews, GitHub collaboration, defect resolution, and the final project demonstration. Responsibilities may overlap where necessary to ensure successful integration and quality improvement.

7. Preliminary Project Schedule

The project is expected to be completed in approximately **4–6 weeks**.

Week	Planned Activities	Expected Deliverables
Week 1	Identify requirements, define project scope, prepare the proposal, establish the GitHub repository, and create initial designs.	Project proposal, initial requirements, repository setup
Week 2	Complete the SRS, database design, interface prototypes, architecture, and quality assurance plan.	SRS, design documents, QA plan
Week 3	Implement parcel registration, tracking number generation, charge calculation, and staff assignment.	Initial working application
Week 4	Implement shipment tracking, delivery attempts, failed delivery handling, and return management. Begin automated testing and defect logging.	Feature-complete prototype, initial test results
Week 5	Perform integration, system, negative, boundary-value, and user acceptance testing. Fix defects and conduct regression testing.	Test execution report, defect log, updated application

Week 6	Calculate quality metrics, consolidate documentation, collect evidence, prepare the final presentation, and conduct the demonstration.	Final application, quality metrics report, final report, presentation

The schedule is preliminary and may be adjusted according to the instructor's deadlines and the team's development progress. If only four weeks are available, design and documentation activities will overlap with implementation, and optional features will be deferred.

8. Expected Outcomes

At the end of the project, the group expects to deliver:

- A functional web-based Parcel Dispatch and Tracking System.
- A structured database for parcel records, staff assignments, and shipment history.
- A documented set of functional and non-functional requirements.
- A software quality assurance plan, test plan, and requirements-to-test traceability matrix.
- Automated and manual test cases with recorded execution results.
- A defect log documenting identified issues, corrections, and retesting.
- A software quality metrics report based on actual collected evidence.
- A GitHub repository containing source code, meaningful commits, issue tracking, and setup instructions.
- A final project report and presentation demonstrating the application and its quality engineering process.

9. Conclusion

The Parcel Dispatch and Tracking System provides a practical solution to common parcel management and delivery tracking problems. Its clearly defined business rules and workflows make it suitable for applying Software Quality Engineering principles throughout the software development lifecycle.

By combining application development with structured testing, defect management, quality metrics, and collaborative version control, the project will demonstrate how software quality can be planned, evaluated, and improved systematically. The proposed scope is appropriate for a three-member group working within an estimated four-to-six-week schedule.

Thank You